TOR AMPLITUDE AGAINST ALL QUASI-COHERENT SHEAVES ON A QUASI-SEPARATED SPACE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete in both directions, closing with the schemes case

Editorial note. The label lemma-tor-qc-qs names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- tor amplitude of an object of the derived category of modules
- etale neighbourhoods U to X with affine U, quasi-compact separated by quasi-separatedness, and pushforward preserving quasi-coherence
- the etale pull-push splitting lemma exhibiting a module as a direct summand of j^*j_*G
- the comparison of the derived category of quasi-coherent modules on a space with the small etale derived category and flatness of the localization functor
- the schemes version of the tor-amplitude criterion

Source reference. Original equation 08H7: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 320–320. Prerequisite pointer invoked at proof line(s) 3408; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 071Q: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 402–402. Prerequisite pointer invoked at proof line(s) 3404; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08H8: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 330–330. Prerequisite pointer invoked at proof line(s) 3416; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08EA: sources/stacks-chow/tags.txt, source lines 10666–10666. Prerequisite pointer invoked at proof line(s) 3422; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0HAP: sources/stacks-chow/tags.txt, source lines 21569–21569. Prerequisite pointer invoked at proof line(s) 3394; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03M9: sources/stacks-chow/tags.txt, source lines 4741–4741. Prerequisite pointer invoked at proof line(s) 3387; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-separated algebraic space over S. Let E be an object of $D_{QCoh}(\mathcal{O}_X)$. Let $a \leq b$. The following are equivalent*

1. *E has tor amplitude in $[a, b]$, and*
2. *for all $\mathcal{F}$ in $QCoh(\mathcal{O}_X)$ we have $H^i(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{F}) = 0$ for $i \notin [a, b]$.*

Proof. It is clear that (1) implies (2). Assume (2). Let $j : U \to X$ be an étale morphism with U affine. As X is quasi-separated, we see that j is both quasi-compact and separated. Hence j_* transforms quasi-coherent modules into quasi-coherent modules (Morphisms of Spaces, Lemma 03M9). Take any quasi-coherent module $\mathcal{G}$ on U. By assumption we see that $H^i(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} j_*\mathcal{G})$ vanishes for $i \notin [a, b]$. Pulling back by the flat morphism j we find that $H^i(E|_U \otimes^{\mathbf{L}}_{\mathcal{O}_U} j^*j_*\mathcal{G})$ vanishes for $i \notin [a, b]$. By Cohomology of Spaces, Lemma 0HAP we see that $\mathcal{G}$ is a direct summand of $j^*j_*\mathcal{G}$. It follows that condition (2) implies the vanishing of $H^i(E|_U \otimes^{\mathbf{L}}_{\mathcal{O}_U} \mathcal{G})$ for $i \notin [a, b]$ for all quasi-coherent $\mathcal{O}_U$-modules $\mathcal{G}$. Since it suffices to prove that $E|_U$ has tor amplitude in $[a, b]$ we reduce to the case where X is representable.

If X is representable by a scheme X_0 then (Lemma 071Q) we can write $E = \epsilon^*E_0$ where E_0 is an object of $D_{QCoh}(\mathcal{O}_{X_0})$ and $\epsilon : X_{étale} \to (X_0)_{Zar}$ is as in (08H7). For every quasi-coherent module $\mathcal{F}_0$ on X_0 the module $\epsilon^*\mathcal{F}_0$ is quasi-coherent on X and

$$H^i(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \epsilon^*\mathcal{F}_0) = \epsilon^* H^i(E_0 \otimes^{\mathbf{L}}_{\mathcal{O}_{X_0}} \mathcal{F}_0)$$

as ϵ is flat (Lemma 08H8). Moreover, the vanishing of these sheaves for $i \notin [a, b]$ implies the same thing for $H^i(E_0 \otimes^{\mathbf{L}}_{\mathcal{O}_{X_0}} \mathcal{F}_0)$ by the same lemma. Thus we've reduced the problem to the case of schemes which is treated in Derived Categories of Schemes, Lemma 08EA. □